

Employee Demo Program

Aim/Objective:

To develop a Java program to demonstrate constructor overloading by creating Employee objects using a default constructor, a constructor with two parameters, and a constructor with all parameters.

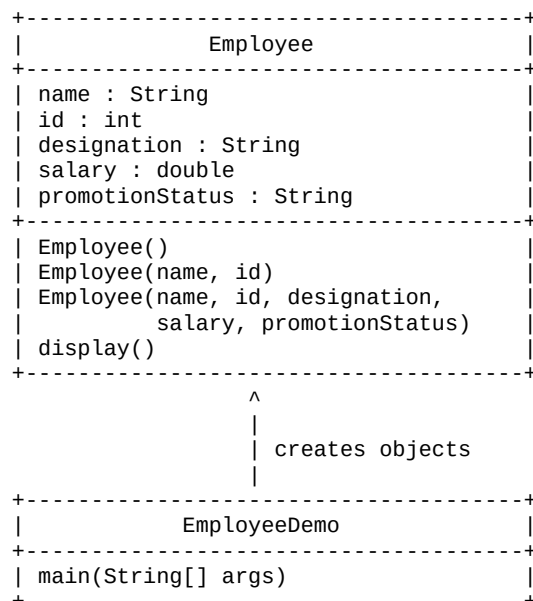
System Requirements:

- Operating System: Windows or any operating system supporting Java.
- Programming Language: Java.
- Java Development Kit (JDK).
- Text Editor or IDE: VS Code / Eclipse / IntelliJ IDEA.
- Command Prompt or Terminal.

Algorithm/Logic Description:

1. Start.
2. Create a class named Employee.
3. Declare name, id, designation, salary, and promotionStatus as data members.
4. Create a default constructor and initialize the employee with default values.
5. Create a constructor with name and id as parameters.
6. Create another constructor with all employee details as parameters.
7. Create a display() method to print the employee details.
8. Create object e1 using the default constructor.
9. Create object e2 using the two-parameter constructor.
10. Create object e3 using the constructor with all parameters.
11. Display the details of all three employees.
12. Stop.

UML Class Diagram:



Source Code:

```
class Employee {
    String name;
```

```

int id;
String designation;
double salary;
String promotionStatus;

// Default Constructor
Employee() {
    name = "Not Assigned";
    id = 0;
    designation = "Trainee";
    salary = 0;
    promotionStatus = "No";
}

// Constructor with two parameters
Employee(String name, int id) {
    this.name = name;
    this.id = id;
    designation = "Developer";
    salary = 30000;
    promotionStatus = "No";
}

// Constructor with all parameters
Employee(String name, int id,
        String designation,
        double salary,
        String promotionStatus) {

    this.name = name;
    this.id = id;
    this.designation = designation;
    this.salary = salary;
    this.promotionStatus = promotionStatus;
}

void display() {
    System.out.println("Name : " + name);
    System.out.println("ID : " + id);
    System.out.println("Designation : " + designation);
    System.out.println("Salary : " + salary);
    System.out.println("Promotion Status : " + promotionStatus);
    System.out.println();
}
}

public class EmployeeDemo {

    public static void main(String[] args) {

        Employee e1 = new Employee();

        Employee e2 = new Employee(
            "Meghana", 101);

        Employee e3 = new Employee(
            "Ravi", 102,
            "Manager", 60000,
            "Yes");

        System.out.println("Employee 1");
        e1.display();

        System.out.println("Employee 2");
        e2.display();

        System.out.println("Employee 3");
        e3.display();
    }
}

```

Compilation and Execution Commands:

Save the file as: EmployeeDemo.java

Compile:

```
javac EmployeeDemo.java
```

Run:

```
java EmployeeDemo
```

Output Screenshot:

```
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> javac EmployeeDemo.java
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> java EmployeeDemo

Employee 1
Name : Not Assigned
ID : 0
Designation : Trainee
Salary : 0.0
Promotion Status : No

Employee 2
Name : Meghana
ID : 101
Designation : Developer
Salary : 30000.0
Promotion Status : No

Employee 3
Name : Mani
ID : 102
Designation : Manager
Salary : 60000.0
Promotion Status : Yes
```

Result:

Thus, the Java program was successfully executed and constructor overloading was demonstrated using the Employee class. The details of three employees were displayed using different constructors.